# Your Presentation Title Here

Course Name or Event

Your Name
Stevens Institute of Technology

April 1, 2026

# Outline

**Introduction**
Background and Motivation

Methods
Proposed Methodology

Results
Experimental Results and Discussion

Conclusion
Summary and Future Directions

# Motivation
## Why this problem matters

**Current challenges:**

- First challenge in your domain
- Second challenge with more detail
- Third challenge highlighting the gap

**Why now?**

- Recent advances make this feasible
- Growing demand from industry

**Key Observation**

Replace this with the central insight that motivates your work.

# Research Questions

**Central Question**

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**Sub-questions:**

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- **RQ2:** Does it generalize across domains?
- **RQ3:** What is the computational cost?

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**Central Question**

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**Sub-questions:**

- **RQ1:** How does the method perform under limited data?
- **RQ2:** Does it generalize across domains?
- **RQ3:** What is the computational cost?

*Each question maps to a specific experiment in the Results section.*

# Contributions

## Contribution 1

### A Novel Method

Brief description of your first contribution.

Achieves **X%** improvement.

Technique Used

*[Published]*

## Contribution 2

### A New Dataset

Brief description of your second contribution.

Contains **N** samples.

Data Collection

*[Published]*

## Contribution 3

### A Framework

Brief description of your third contribution.

Open-source release.

System Design

*[Proposed]*

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# Comparison of Approaches

## Why existing methods fall short

| Property | Method A | Method B | Ours |
|---|---|---|---|
| Scalable | ✓ | ✗ | ✓ |
| Interpretable | ✗ | ✓ | ✓ |
| Low latency | ✓ | ✓ | ✓ |
| Domain-agnostic | ✗ | ✗ | ✓ |

**Advantage**

Our approach is the first to satisfy all four desirable properties simultaneously.

# Proposed Pipeline
## Three-stage processing

**Stage 1: Data Preprocessing**

1. Collect raw data from source
2. Clean and normalize
3. Split into train/val/test

**Stage 2: Model Training**

1. Fine-tune pre-trained model
2. Hyperparameter search
3. Early stopping on validation set

**Stage 3: Evaluation**

1. Run on held-out test set
2. Compute metrics (P, R, F1)
3. Statistical significance tests

**Key Insight**

Pre-training captures domain knowledge; fine-tuning adapts with minimal supervision.

# Block Styles

## Three built-in block types

### Standard Block

This is a standard block with the Stevens red header. Use it for general information, definitions, or neutral content.

### Alert Block

This is an alert block. Use it for warnings, limitations, or important caveats that readers should be aware of.

### Example Block

This is an example block with a green header. Use it for examples, case studies, or positive results.

# Mathematical Formulation

**Objective function:**

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N} \log p_\theta(y_i \mid x_i) + \lambda \|\theta\|_2^2$$

**Where:**

- $\theta$ — model parameters
- $(x_i, y_i)$ — input-output pairs
- $\lambda$ — regularization strength

**Note**

The regularization term prevents overfitting when training data is limited.

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# Main Results

## Performance comparison on benchmark dataset

| Method | Precision | Recall | F1 | Acc. |
|---|---|---|---|---|
| Baseline A (100% data) | .834 | .821 | .827 | 85.3 |
| Baseline B (100% data) | .856 | .847 | .851 | 87.1 |
| Ours (10% data) | .871 | .863 | **.867** | **88.4** |
| Ours (100% data) | .912 | .905 | **.908** | **91.7** |

**Key Finding**

With only **10% of training data**, our method already outperforms baselines using full data.

# Ablation Study
## Contribution of each component

| Configuration | F1 Score | $\Delta$ |
|---|---|---|
| Full model | **0.908** | — |
| – Component A | 0.881 | −0.027 |
| – Component B | 0.873 | −0.035 |
| – Component C | 0.854 | −0.054 |
| – All components | 0.827 | −0.081 |

**Key insight:** Component C contributes the most (−5.4 F1), confirming our hypothesis.

# Results Summary

**Findings:**

- Outperforms all baselines on F1 and accuracy
- Robust under low-resource conditions (10% data)
- Component C is the most impactful addition

**Limitations:**

- Higher inference latency than Baseline A
- Not yet tested on multilingual data

**Takeaway**

Strong performance with minimal supervision demonstrates practical applicability.

**Next steps:**

- Multilingual evaluation
- Deployment study

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# Summary of Contributions

| Contribution | Method | Challenge | Result |
|---|---|---|---|
| **Contribution 1** | Technique A | Problem X | 90% improvement |
| **Contribution 2** | Technique B | Problem Y | 50× scaling |
| **Contribution 3** | Technique C | Problem Z | Novel framework |

**Central Message**

Replace this with your one-sentence thesis or takeaway.

**Thank you for listening!**

**Questions?**